

An observation on T_A :

$$T_A(q) = q^2 \sigma_1(V_{(2,2,1)}) \text{ at } S_5$$

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Abstract

A short addendum to the May 7 writeup `2026-05-07-trace-identity.tex`, which reduced the surprising trace identity $D(q) = \text{cross}_{(3,1,1) \rightarrow (3,2)}(q)$ to a residual identity $(1+q)(T_A + T_B) = v \text{tr}(P_{(3,1,1)} \Pi^{S_5} P_{(3,2)} M_5 R'_5)$, with the explicit values $T_A = q^6(1+q)^2$ and $T_B = q^5((1+q)^4 + 2q(1+q)^2)$. The May 7 writeup left T_A 's structural origin open. Here we observe a clean factorization:

$$T_A(q) = q^2 \sigma_1(V_{(2,2,1)})|_{S_5}$$

where $\sigma_1(V_{(2,2,1)})|_{S_5} = q^4(1+q)^2$ is the σ_1 trace of $V_{(2,2,1)}$ as an irreducible $H(S_5)$ -cell module. We do not prove this structurally, but the factorization is striking: an $H(S_6)$ -trace involving the KL projection onto $V_{(2,2,1)}$ -summand, weighted by an extra 4-letter Bott–Samelson factor R'_5 , equals (up to the rigid factor q^2) the standalone σ_1 at S_5 of the S_5 -cell. This suggests a representation-theoretic or branching-functorial source of the identity.

1 Setup

Conventions follow `2026-05-07-trace-identity.tex`: the C_w KL basis on $V_{(3,2,1)} \subset H_q(S_6)$, with $T_i + 1 = (1+q)D_i + vM_i$ where D_i records left *ascents* ($s_i \notin D_L(w)$). The staircase factorization $\Pi_q = \Pi^{S_5} \cdot (T_5 + 1) \cdot R'_5$ has $\Pi^{S_5}, R'_5 \in H(S_5)$.

The S_5 -branching $V_{(3,2,1)} \downarrow S_5 = V_{(3,2)} \oplus V_{(3,1,1)} \oplus V_{(2,2,1)}$ and the diagonal projector $P_{(2,2,1)}$ onto the KL block $\{e_5, e_7, e_{11}, e_{13}, e_{15}\}$ are as in `s5_block_decomp.txt`. Define

$$T_A := \text{tr}_{V_{(3,2,1)}}(\Pi^{S_5} P_{(2,2,1)} R'_5).$$

The May 7 writeup verifies $T_A = q^6(1+q)^2 = q^6 + 2q^7 + q^8$ by Lagrange interpolation.

2 The factorization

Theorem 1 (Numerically verified).

$$T_A(q) = q^2 \sigma_1(V_{(2,2,1)})|_{S_5}.$$

Verification. From `s5_block_decomp.txt`:

$$\sigma_1(V_{(2,2,1)})|_{S_5} = q^4 + 2q^5 + q^6 = q^4(1+q)^2.$$

Multiplying by q^2 :

$$q^2 \cdot q^4(1+q)^2 = q^6(1+q)^2 = q^6 + 2q^7 + q^8 = T_A(q). \quad \square$$

Remark 2 (Why this is striking). The two polynomials are computed in different settings:

- T_A is the $H(S_6)$ -trace on $V_{(3,2,1)}$ (a 16-dim KL-basis module), with the S_5 -staircase Bott–Samelson sandwiched between a KL-diagonal projection $P_{(2,2,1)}$ and an extra 4-letter R'_5 .
- $\sigma_1(V_{(2,2,1)})|_{S_5}$ is the $H(S_5)$ -trace on the irreducible cell module $V_{(2,2,1)}$ of S_5 (a 5-dim KL-basis module), with the S_5 -staircase Bott–Samelson *alone*.

The identification of the two KL bases is non-trivial. The 5-dim space spanned by $\{C_{w_5}, C_{w_7}, C_{w_{11}}, C_{w_{13}}, C_{w_{15}}\} \subset V_{(3,2,1)}$ (in the $H(S_6)$ KL basis) is isomorphic, as an $H(S_5)$ -module, to $V_{(2,2,1)}$ at S_5 , but the explicit basis elements differ from the $H(S_5)$ KL basis of $V_{(2,2,1)}$ standalone. So Theorem ?? is a non-trivial coincidence between two independently-defined polynomial traces.

3 Implications

If Theorem ?? can be lifted to a structural identity, the May 7 residual reduction tightens substantially. Recall (May 7, eq. (??)) the residual identity is $(1+q)(T_A+T_B) = v \operatorname{tr}(P_{(3,1,1)} \Pi^{S_5} P_{(3,2)} M_5 R'_5)$. Substituting the conjectured factorization:

$$(1+q) q^2 \sigma_1(V_{(2,2,1)})|_{S_5} + (1+q) T_B = v \operatorname{tr}(P_{(3,1,1)} \Pi^{S_5} P_{(3,2)} M_5 R'_5).$$

The first term is now wholly intrinsic to $V_{(2,2,1)}$ at S_5 . If a parallel factorization $T_B = (\text{something}) \cdot \sigma_1(V_?)|_{S_5}$ holds, the structural origin of the identity becomes clearer.

Searching for an analogous T_B factorization

Recall $T_B = q^5((1+q)^4 + 2q(1+q)^2) = q^5 + 6q^6 + 10q^7 + 6q^8 + q^9$. The factor $(1+q)^4 + 2q(1+q)^2$ is a B_4 -atom of multiplicity vector $(1, 2, 0)$.

The natural candidate is $\sigma_1(V_{(3,1,1)})|_{S_5}$:

$$\sigma_1(V_{(3,1,1)})|_{S_5} = q^3 + 6q^4 + 10q^5 + 6q^6 + q^7 = q^3((1+q)^4 + 2q(1+q)^2).$$

Multiplying by q^2 :

$$q^2 \cdot \sigma_1(V_{(3,1,1)})|_{S_5} = q^5 + 6q^6 + 10q^7 + 6q^8 + q^9 = T_B(q).$$

Theorem 3 (Numerically verified).

$$T_B(q) = q^2 \sigma_1(V_{(3,1,1)})|_{S_5}.$$

Verification. From `s5_block_decomp.txt`: $\sigma_1(V_{(3,1,1)})|_{S_5} = q^3 + 6q^4 + 10q^5 + 6q^6 + q^7 = q^3(1+q)^4 + 2q^4(1+q)^2$, factored as $q^3 \cdot ((1+q)^4 + 2q(1+q)^2)$. Multiplying by q^2 yields the claimed value of T_B . $\square$

This is a second clean factorization. Both summands of the May 7 residual decomposition are q^2 times an intrinsic S_5 -cell σ_1 :

Corollary 4 (Structural form of the residual). *The May 7 residual identity (??) is equivalent to*

$$v \operatorname{tr}(P_{(3,1,1)} \Pi^{S_5} P_{(3,2)} M_5 R'_5) = (1+q) q^2 [\sigma_1(V_{(2,2,1)})|_{S_5} + \sigma_1(V_{(3,1,1)})|_{S_5}]. \quad (1)$$

Proof. $T_A + T_B = q^2 \sigma_1(V_{(2,2,1)})|_{S_5} + q^2 \sigma_1(V_{(3,1,1)})|_{S_5}$ by Theorems ?? and ??. Substitute into (??). $\square$

What $V_{(3,2)}$ does *not* contribute

Observe that the third summand $\sigma_1(V_{(3,2)})|_{S_5}$ is *absent* from the right-hand side of (??). From `main_output.txt`, $\sigma_1(V_{(3,2)})|_{S_5} = q^2(1+q)^2(q^4 + 9q^3 + 19q^2 + 9q + 1)$, which is the largest of the three by both degree and $q = 1$ value ($\sigma_1(V_{(3,2)})|_{S_5}|_{q=1} = 156$, vs. $\sigma_1(V_{(3,1,1)})|_{q=1} = 24$, $\sigma_1(V_{(2,2,1)})|_{q=1} = 4$).

The structural meaning: $V_{(3,2)}$'s role in the trace identity is fundamentally different from the other two. Recall $V_{(3,2)}$ is precisely the summand into which all non-zero KL cross-flow lands (cross $_{\mu \rightarrow \nu} = 0$ for $\nu \neq (3,2)$). So $V_{(3,2)}$ acts as the “sink” summand, while $V_{(3,1,1)}$ and $V_{(2,2,1)}$ are the “sources”. Corollary ?? says: the LHS of the residual (the M-step cross trace, which IS the cross-flow into $V_{(3,2)}$) is encoded by the source-summand σ_1 's, weighted by $q^2(1+q)$.

4 The structural conjecture

The two factorizations cry out for a representation-theoretic explanation.

Conjecture 5 (Branching factorization). *For each $\mu \in \{(2,2,1), (3,1,1)\}$ in $V_{(3,2,1)} \downarrow S_5$,*

$$\mathrm{tr}_{V_{(3,2,1)}}(\Pi^{S_5} P_\mu^{\mathrm{KL}} R'_5) = q^2 \sigma_1(V_\mu)|_{S_5},$$

where P_μ^{KL} is the diagonal projection in the $H(S_6)$ KL basis onto the V_μ -block of vertices.

The q^2 factor is the intriguing common feature. Two candidate explanations:

- **Factorization through an intermediate operator.** Perhaps R'_5 acts on the V_μ -summand of $V_{(3,2,1)} \downarrow S_5$ (under the $H(S_5)$ -action) as $q^2 \cdot I + (\text{trace-zero-on-}V_\mu\text{-correction})$. The trace of $\Pi^{S_5} \cdot R'_5$ then collapses to $q^2 \cdot \sigma_1(V_\mu)$. This would require checking R'_5 explicitly on each V_μ .
- **Branching-functorial origin.** The Bott–Samelson product $\Pi^{S_5} R'_5$ (a 14-letter word for an element in S_5 which has length 10) decomposes into $\Pi^{S_5} \cdot R'_5$ where the second factor is “redundant” in the reduced-word sense. Under the S_5 -branching, perhaps the redundant factor R'_5 contributes a clean q^2 to each summand of $V_{(3,2,1)} \downarrow S_5$ *except* the largest summand $V_{(3,2)}$, where the contribution is more complex (cross-flow).

5 Honest assessment

What this paper achieves.

1. The two clean factorizations $T_A = q^2 \sigma_1(V_{(2,2,1)})|_{S_5}$ and $T_B = q^2 \sigma_1(V_{(3,1,1)})|_{S_5}$, both verified numerically.
2. A structural rephrasing of the May 7 residual identity (Corollary ??) as a sum of intrinsic S_5 -cell σ_1 values, multiplied by $q^2(1+q)$.
3. The branching factorization conjecture (Conjecture ??), now in clean “two-summand” form.

What this paper does *not* achieve.

1. A proof of Conjecture ??.
2. A structural proof of the May 7 residual identity (??) (i.e., the original $D = \text{cross}$).
3. A representation-theoretic interpretation of the q^2 factor.

Status. The May 7 reduction has been refined: instead of two opaque polynomial values T_A, T_B , we have two clean factorizations. The residual gap is now: *prove the branching factorization conjecture*. This is more constrained than the original problem (it asks why two specific Bott–Samelson sub-traces equal q^2 times intrinsic σ_1 values), and possibly more amenable to attack via W -graph theory or branching coefficients.

Computational note. All polynomial values used are taken from `~/projects/scratch/2026-05-06-paths/main` and `s5_block_decomp.txt`. The factorizations $T_A = q^2 \sigma_1(V_{(2,2,1)})$ and $T_B = q^2 \sigma_1(V_{(3,1,1)})$ are verified by direct polynomial multiplication.